

Task 04 — Temperature Experiment

1.Objective

The objective of this experiment is to observe how changing the temperature parameter affects the variation and creativity of LLM-generated responses. The same prompt was tested at temperatures 0, 0.7, and 1.0, with three runs performed at each temperature.

2. Experimental Setup

Parameter	Value
Model	Gemini 3.1 Flash-Lite
Prompt	Same prompt for all experiments
Temperatures	0, 0.7, 1.0
Runs per temperature	3
Total runs	9

3. Prompt Used

Write a short description of a rainy evening in a busy city. Describe the atmosphere, surroundings, and emotions in 80–100 words.

4. Temperature 0.0 — 3 Outputs

Temperature	Output
0.0	Neon lights bleed across slick asphalt, turning the city streets into a shimmering, kaleidoscopic river. Rain drums a relentless rhythm against glass towers, blurring the frantic movement of umbrellas into a sea of shifting shadows. The air tastes of ozone and exhaust, thick with the hum of idling engines and distant sirens. Amidst the chaos, a quiet melancholy settles; the downpour acts as a veil isolating each passerby in their own private world. It is a symphony of urban solitude, where the cold dampness of the night feels strangely comforting against the relentless, glowing pulse.
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5. Temperature 0.7 — 3 Outputs

Temperature	Output
0.7	Neon signs bleed into slick, rain-drenched pavement, turning the busy avenue into a kaleidoscope of reflected light. Umbrellas collide in a rhythmic dance as commuters rush through the downpour, their figures blurred by the mist. The air smells of wet asphalt and ozone, heavy with the relentless hum of idling taxis. Amidst the urban cacophony, a profound sense of isolation settles in; the city feels vast and indifferent. Yet, there is a melancholic beauty in the chaos—a quiet, shivering intimacy found in watching the world blur behind a veil of silver rain.
0.7	Neon signs fracture against slick asphalt, painting the city in bleeding streaks of electric blue and crimson. Rain drums a relentless rhythm against glass storefronts, drowning out the frantic symphony of honking taxis and hurried footsteps. Commuters huddle under umbrellas, heads bowed against the biting wind, their silhouettes blurring into the grey, misty haze. There is a melancholy comfort in the chaos; the air tastes of wet concrete and ozone. Amidst the relentless rush, the city feels both suffocating and strangely intimate, a vast, shivering organism finding shelter beneath a weeping, ink-black sky.
0.7	Neon signs bleed into slick, rain-drenched asphalt, casting vibrant ribbons of electric blue and crimson across the city. The air hangs heavy with the scent of ozone and exhaust, while the rhythmic hiss of tires on wet pavement serves as the evening's restless heartbeat. Pedestrians vanish into a sea of bobbing umbrellas, their hurried footsteps echoing against towering glass monoliths. Amidst the relentless downpour, a profound sense of solitude settles in. It is a beautiful, melancholic chaos—a world blurred at the edges, where the frantic pulse of the metropolis feels strangely intimate and quiet.

6. Temperature 1.0 — 3 Outputs

Temperature	Output
1.0	Neon lights fracture across rain-slicked asphalt, painting the city in bleeding streaks of electric blue and crimson. The air is thick with the scent of wet concrete and ozone, punctuated by the rhythmic hiss of tires cutting through deep puddles. Pedestrians hunched under translucent umbrellas rush past, their faces obscured by the shadows and the gray, unrelenting downpour. Despite the frenetic motion, a strange, melancholy stillness settles in the gaps between honking horns. It is a lonely, cinematic solitude, where the frantic pulse of the metropolis softens into a soothing, monochromatic hum against the dark.
1.0	Neon lights bleed across the slick, obsidian pavement, transforming the city into a shivering kaleidoscope of blurred color. Rain lashes against glass, drowning the distant hum of traffic in a steady, rhythmic percussion. Commuters scurry like shadows beneath arched umbrellas, their faces tucked against the biting wind. The air tastes of wet asphalt and ozone, heavy with a melancholic solitude despite the suffocating crush of humanity. It is a cinematic isolation; here, amidst the towering steel and frantic motion, the downpour offers a brief, quiet sanctuary for those lost in the gray urban rush.
1.0	Neon lights bleed into the slick, black asphalt, transforming the city streets into a kaleidoscope of shattered reflections. Rain drumbeats against metal awnings, a rhythmic hiss that drowns out the distant hum of traffic. Pedestrians dash through the downpour, hunched under umbrellas like migrating beetles, their faces obscured by the gray, swirling mist. There is a strange, melancholy comfort in the chaos; the damp air smells of ozone and exhaust. Caught between the relentless glow of skyscrapers and the lonely pulse of the storm, the city feels both suffocatingly crowded and beautifully, hauntingly isolated.

7. Comparison / Variance Analysis

Temperature	Run-to-Run Variation	Observed Variance	Key Observation
0.0	None	Very Low	All three outputs were exactly identical.
0.7	Noticeable	Moderate	Outputs remained consistent in theme but varied in wording, imagery, and descriptions.
1.0	High	High	Outputs showed greater creative variation in imagery, metaphors, and sentence construction.

Increasing temperature from 0 to 0.7 and then to 1.0 increased the observed variation in the generated responses while the overall topic remained consistent.

8. Recommended Temperature by Use Case

Use Case	Recommended Temperature	Justification
Support Bot	0.0	Support systems generally benefit from predictable and consistent responses. A lower temperature reduces unnecessary variation and helps maintain stable responses to similar queries.
Code Generator	0.0	Code generation benefits from consistency and precision. Lower variation can reduce unnecessary changes in generated code and make outputs more predictable.
Marketing Copy	1.0	Marketing content can benefit from creative variation. A higher temperature can produce more diverse wording, metaphors, and ideas, which can be useful when generating creative copy.

Why not use 0.7?

A temperature of **0.7** provides a balance between consistency and creativity. It could therefore be suitable when an application requires some variation while still maintaining relatively controlled outputs.

9. Conclusion

This experiment demonstrated that temperature has a noticeable effect on the variation of LLM-generated text. At temperature 0, all three responses were identical, while temperatures 0.7 and 1.0 produced progressively greater variation in wording, imagery, and creative expression. Therefore, lower temperatures are more suitable for applications requiring consistency and predictability, while higher temperatures can be useful for creative applications where greater variation is desirable.